19 Current Electricity

Electric Current

1. **Assertion (A):** Current is a vector quantity.
Reason (R): Current has magnitude as well as direction.
(*a*) Assertion (A) is false but Reason (R) is true
(*b*) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(*c*) Assertion (A) is true but Reason (R) is false
(*d*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

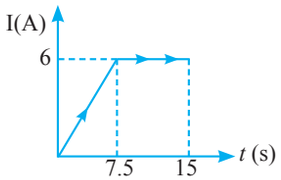
2. A current of 0.6 A is drawn by an electric bulb for 10 minutes. Which one of the following is the amount of electric charge that flows through the circuit?
(*a*) 6 C (*b*) 0.6 C (*c*) 360 C (*d*) 36 C

3. If the amount of charge passing through the cell in 4 minutes is 480 C then calculate the current passing through the cell?

4. The current in a wire varies with time according to the equation $I = 4 + 2t$, where I is in ampere and t is in sec. The quantity of charge which has passed through a cross-section of the wire during the time $t = 2$ sec to, $t = 6$ sec will be:
(*a*) 60 coulomb (*b*) 24 coulomb
(*c*) 48 coulomb (*d*) 30 coulomb

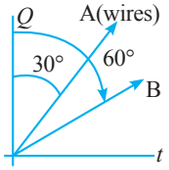
5. If charge $Q = t^2 + 2t - 4$ is flowing through wire then find current at $t = 2$ sec and in 2 sec.

6. The graph of current versus time in a wire is given below, If charge flows through the wire in 7.5s is q_1 and that flows through the wire in 15s is q_2, then ratio $q_1 : q_2$ is



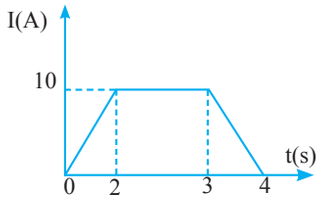
(*a*) 2 : 1 (*b*) 1 : 2
(*c*) 4 : 1 (*d*) 1 : 3

7. Charge Flowing through wire A & B as given in graph then find ratio of current through A & B

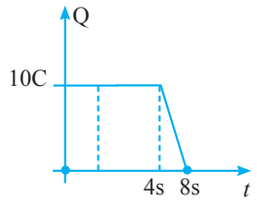


8. Through a given cross section n_1 electrons per second are passing from left to right and n_2 protons per second are passing from right to left simultaneously. The electric current through that cross section is (e = electronic charge)
(*a*) $(n_1 + n_2)\, e$ towards left (*b*) $(n_2 - n_1)\, e$ towards right
(*c*) $(n_1^2 + n_2^2)\, e$ towards left (*d*) $(n_2^2 + n_1^2)$ towards right

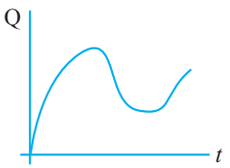
9. Find average current in 4 sec?



10. Find average current in 8 sec?



11. Find how many time current is zero in given graph?



12. Match column-I with Column-II:

Column-I		Column-I	
i	If $Q = t^2 + 2t - 4$, I in 2 seconds	A	6 A
ii	If $Q = t^2 + 2t - 4$, I at 2 seconds	B	28 A
iii	If $I = 3t^2 + 4$, average current in 2 seconds	C	4 A
iv	If $Q = 5t^2 - 2t$, $I_{average}$ = ?B/W $t_1 = 2$s to $t_2 = 4$s	D	8 A

(*a*) (i)-D, (ii)-A, (iii)-B, (iv)-C
(*b*) (i)-C, (ii)-A, (iii)-D, (iv)-B
(*c*) (i)-A, (ii)-B, (iii)-C, (iv)-A
(*d*) (i)-B, (ii)-C, (iii)-A, (iv)-C

13. x is the current if electron is moving on circular path with frequency f and y is the current if e is moving with speed v in circle of radius R. Find x / y :

(*a*) $\frac{2\pi RF}{v}$ (*b*) $\frac{2\pi RF}{f}$ (*c*) $\frac{ef}{2\pi RF}$ (*d*) $\frac{\pi^2 ef}{2Rv}$

14. A current of 5A exists in a 10 Ω resistance for 4 minutes. The number of electrons and charge (in coulombs) passing through any section of the resistor in this time are

(*a*) 75×10^{20}, 600 C (*b*) 75×10^{21}, 600 C

(*c*) 75×10^{20} , 1200 C (*d*) 75×10^{19}, 1200 C

15. **Statement I:** A conductor carrying electric current becomes electrically charged.

Statement II: A conductor carrying electric current contains same number of positive and negative charges and is electrically neutral.

Drift Velocity and Conductivity

16. **Assertion (A) :** Electric field outside the conductor connected to battery of V volt is zero.

Reason (R) : Conductor remains neutral as number of incoming electron is equal to number of outgoing electrons.

(*a*) Assertion (A) is false but Reason (R) is true

(*b*) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(*c*) Assertion (A) is true but Reason (R) is false

(*d*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

17. Two wires A and B of the same material, having radii in the ratio 1 : 2 carry currents in the ratio 4 : 1. The ratio of drift speed of electrons in A and B is

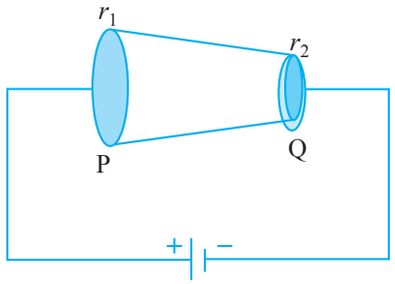
(*a*) 16 : 1 (*b*) 1 : 16 (*c*) 1 : 4 (*d*) 4 : 1

18. The drift velocity of electrons for a conductor connected in an electrical circuit is V_d The conductor is now replaced by another conductor with same material and same length but double the area of cross-section. The applied voltage remains same. The new drift velocity of electrons will be

(*a*) $V_d/4$ (*b*) V_d (*c*) $2V_d$ (*d*) $V_d/2$

19. In given figure, a batter of emf is connected across a conductor PQ of length l and different area of cross-sections having radii r_1 and $r_2 (r_2 < r_1)$.

Choose correct options as one moves from P to Q .



(*a*) Drift velocity of e^- increases

(*b*) Electric field decreases

(*c*) Electron current decreases

(*d*) All of these

20. Across a metallic conductor of non uniform cross section, a constant potential difference is applied. The quantity which remains constant along the conductor is

(*a*) current (*b*) drift velocity

(*c*) electric field (*d*) current density

21. Three wires of same material are connected in parallel to a source of emf. The length ratio of the wires is 1 : 2 : 3 and the ratio of their area of cross section is 2 : 4 : 1

Then match the Ratio of quantities in Column - I with respective answer in Column - II

	Column-I		Column-II
A.	Resistance	(i)	1 : 1 : 1
B.	Current	(ii)	6 : 6 : 1
C.	Power	(iii)	1 : 6 : 6
D.	Conductivity	(iv)	1 : 1 : 6

(*a*) A-(i), B-(iv), C-(iii), D-(ii)

(*b*) A-(iv), B-(ii), C-(ii), D-(i)

(*c*) A-(iii), B-(i), C-(ii), D-(iv)

(*d*) A-(iv), B-(iii), C-(i), D-(ii)

Mobility

22. **Statement-I:** In an isolated conductor, velocity of e^- is zero hence. They are in rest condition as no electric field is applied.

Statement-II: Drift velocity becomes double and hence due to this, mobility of electron will also gets doubled.

(*a*) Both statements false

(*b*) Both statements true

(*c*) Statement-I is false and Statement-II is true

(*d*) Statement-I is true and Statement-II is false

23. Is mobility depends on electric field?

24. Compare mobility of electron and proton?

25. Ratio for mobility of deuteron, proton and a-particle is 2 : x : 2. Find value of x.

26. A charged particle having drift velocity of 7.5×10^{-4} m s^{-1} in an electric field of 3 x 10^{-10} V m^{-1}, has a mobility in m^2 $V^{-1}s^{-1}$ of

(*a*) 2.25×10^{15} (*b*) 2.5×10^{6}

(*c*) 2.5×10^{-6} (*d*) 2.25×10^{-15}

27. The dimensional formula of mobility is

(*a*) $[M^{-1}L^0T^2A^1]$ (*b*) $[M^{-1}L^0T^1A^1]$

(*c*) $[M^{-1}L^{-1}T^2A^1]$ (*d*) $[[M^{-1}L^{-1}T^1A^1]$

28. Current density $\vec{j}$ at an area is $\vec{A} = (2\hat{i} + 3\hat{i})$ mm^2 is $\vec{j} = (3\hat{i} + 4\hat{i})$ A/ m^2. Current through the area is

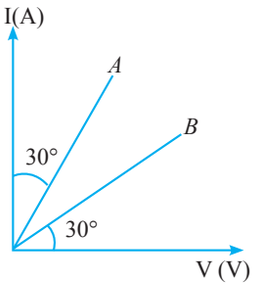
(*a*) 9 μA (*b*) Zero (*c*) 18 μA (*d*) 12 μA

Resistance and Resistivity

29. Three copper wires of length and cress-sectional areas are [l, A], $\left[2l, \frac{A}{2}\right]$, and $\left[\frac{l}{2}, 2A\right]$. Resistivity is maximum is

(a) Wire of cross-sectional area 2A
(b) Wire of cross-sectional area A/2
(c) Wire of cross-sectional area A.
(d) Same in all three cases.

30. Figure shows graph between l and V for two conductor A and B. Their respective resistance are in the ratio.



(a) 1 : 1 (b) 1 : 3 (c) 3 : 1 (d) 1 : 2

31. A certain piece of copper is to be shaped into a conductor of minimum resistance. Its length and diameter should respectively be

(a) L, D (b) $2l, \frac{D}{2}$ (c) $\frac{L}{2}, 2D$ (d) $L, \frac{D}{2}$

32. A wire of resistance x ohm is drawn out so that its length is increased to twice its original length, and its mew resistance becomes 20 Ω, then x will be

(a) 5 Ω (b) 10 Ω (c) 15 Ω (d) 20 Ω

33. A 1m long wire is broken into two unequal parts x and y. The x part of wire is stretched into another wire W. Length of W is twice the length of x and resistance of W is twice that of y. Find ratio of length of x and y.

(a) 1: 4 (b) 1 : 2 (c) 4 : 1 (d) 2 : 1

34. An electric cable of copper has just one wire of radius 9 mm. Its resistance is 14 Ω. If this single copper wire of cable is replaced by seven identical well insulated copper wire each of radius 3 mm connected in parallel, then the new resistance of combination will be:

(a) 9 Ω (b) 18 Ω (c) 28 Ω (d) 126 Ω

35. The resistance of a wire is 'R' ohm. If it is melted and stretched to 'n' times its original length its new resistance will be

(a) $\frac{R}{n}$ (b) $\frac{R}{n^2}$ (c) n^2R (d) nR

36. A wire of a certain material is stretched slowly by ten percent. Its new resistance and specific resistance become respectively

(a) both remain the same (b) 1.1 times, 1.1 times
(c) 1.2 times, 1.1 times (d) 1.21 times, same

37. The masses of the wire of copper is in the ratio of 1 : 3 : 5 and their lengths are in the ratio 5 : 3 : 1. The ratio of their electrical resistance is

(a) 1 : 3 : 5 (b) 5 : 3 : 1
(c) 1 : 25 : 125 (d) 125 : 15 : 1

38. The electric resistance of a certain wire of iron is R. If its length and radius are both doubled, then

(a) The resistance will be doubled and the specific resistance will be halved
(b) The resistance will be halved and the specific resistance will remain unchanged
(c) The resistance will be halved and the specific resistance will the doubled
(d) The resistance and the specific resistance. will both remain unchanged

39. If a copper wire stretched to make its radius decrease by 0.1%, then the percentage increase in resistance is nearly

(a) 0.1% (b) 0.8%
(c) 0.4% (d) 0.2%

40. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Alloys such as constantan and manganin are used in making standard resistance coils.

Reason (R): Constantan and manganin have very small value of temperature coefficient of resistance.

In the light of the above statements, choose the correct answer from the options given below.

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(b) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(c) Assertion (A) is true but Reason (R) is false
(d) Assertion (A) is false but Reason (R) is true

41. The solids which have the negative temperature coefficient of resistance are

(a) semiconductors only
(b) insulators and semiconductors
(c) metals
(d) insulators only

42. **Statement I:** Constant potential difference is applied across a conductor. If temperature of conductor is increased, the drift speed of electrons will decrease.

Statement II: Resistivity of a conductor increases with increase in temperature.

43. Resistance wire used in resistance boxes have

(a) maximum resistance
(b) minimum resistance
(c) minimum value of thermal coefficient of resistance
(d) maximum value of thermal coefficient of resistance

Ohm's law

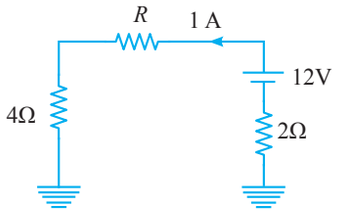
44. The potential difference $(V_A - V_B)$ between the points A and B in the given figure is



(*a*) –3 V (*b*) +3 V

(*c*) +6 V (*d*) +9 V

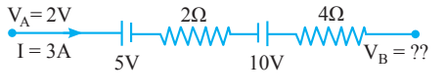
45. Value of the resistance R in the figure is



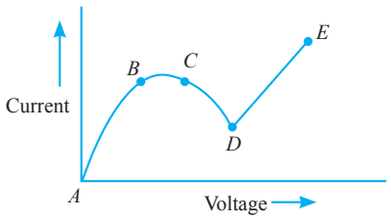
(*a*) 6 Ω (*b*) 8 Ω

(*c*) 10 Ω (*d*) 12 Ω

46. Find potential at point B



47. From the graph between current (I) and voltage (V) is shown. Identity the portion corresponding to negative resistance



(*a*) CD (*b*) DE

(*c*) AB (*d*) BC

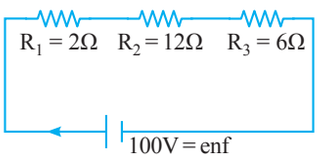
48. A 6 volt battery is connected to the terminals of a three meter long wire of uniform thickness and resistance of 100 ohm. The difference of potential between two points on the wire separated by a distance of 50 cm will be

(*a*) 2 volt (*b*) 3 volt

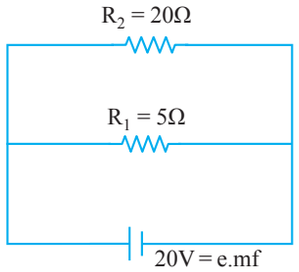
(*c*) 1 volt (*d*) 1.5 volt

Series and Parallel Combination of Resistor

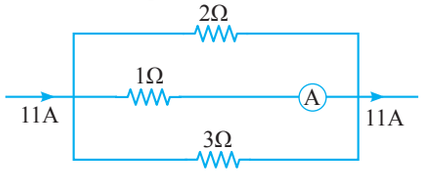
49. Find potential drop across each resistance:-



50. Find the current in each resistor.

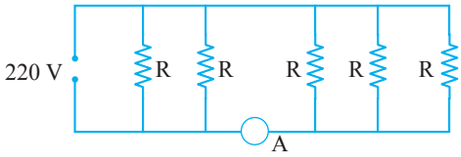


51. The ammeter reading in the circuit below is



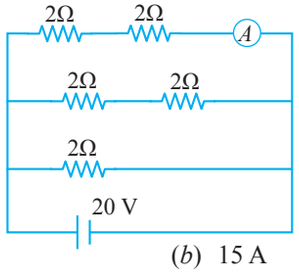
(*a*) 2 A (*b*) 3 A (*c*) 6 A (*d*) 5 A

52. Five identical lamps each of resistance R = 1100 Ω are connected to 220 V as shown in the figure. The reading of ideal ammeter A is



(*a*) $\frac{1}{5}A$ (*b*) $\frac{2}{5}A$ (*c*) $\frac{3}{5}A$ (*d*) 1 A

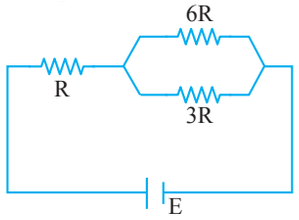
53. The reading of the ammeter in the circuit below is



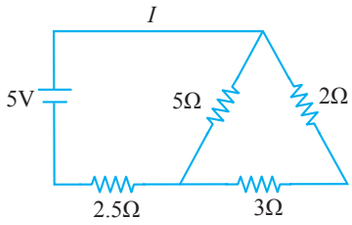
(*a*) 5 A (*b*) 15 A

(*c*) 20 A (*d*) 25 A

54. Find potential drop across '3R'

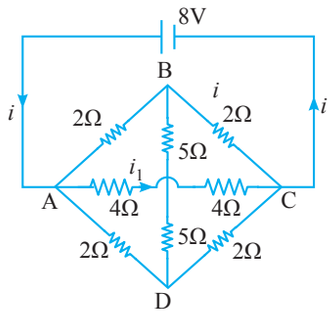


55. The current I through the cell in the network shown is



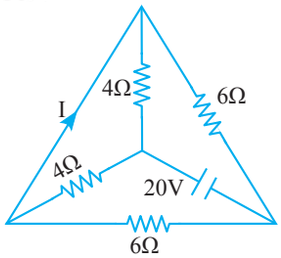
(*a*) 3 A (*b*) 2 A (*c*) 1 A (*d*) 4 A

56. The value of current i_1 flowing from A to C in circuit diagram is:



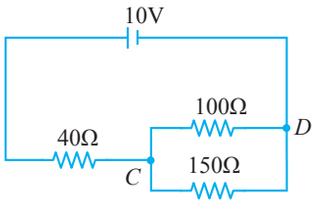
(*a*) 1 A (*b*) 2 A (*c*) 4 A (*d*) 5 A

57. The value of I is :



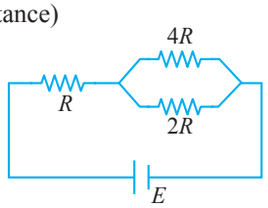
(*a*) 4 A (*b*) 0 A
(*c*) 2 A (*d*) 3 A

58. In the network shown below, potential difference across *CD* is



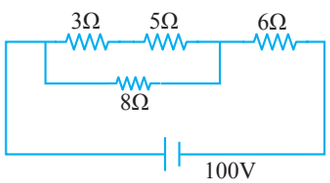
(*a*) 4 V (*b*) 6 V
(*c*) 10 V (*d*) 5 V

59. In a network as shown in the figure, the potential difference across the resistance $2R$ is (the cell has emf E volt and no internal resistance)



(*a*) 2E (*b*) 4E/7
(*c*) E/7 (*d*) E

60. Find current and potential drop in 3Ω



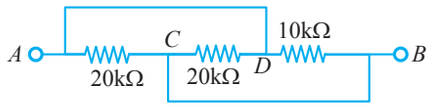
61. Two metallic wires of identical dimensions are connected in series. If σ_1 and σ_2 are the Conductivities of the these wires respectively, the effective conductivity of the combination is:

(*a*) $\frac{\sigma_1\sigma_2}{\sigma_1+\sigma_2}$ (*b*) $\frac{2\sigma_1\sigma_2}{\sigma_1+\sigma_2}$

(*c*) $\frac{\sigma_1+\sigma_2}{2\sigma_1\sigma_2}$ (*d*) $\frac{\sigma_1+\sigma_2}{\sigma_1\sigma_2}$

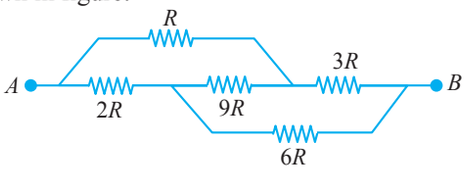
Equivalent Resistance

62. The equivalent resistance between A and B as shown in figure is:



(*a*) 5 kΩ (*b*) 30 kΩ (*c*) 10 kΩ (*d*) 20 kΩ

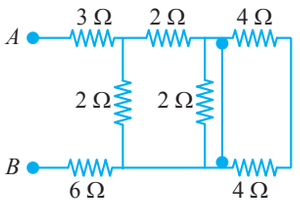
63. The equivalent resistance between *A* and *B* of the network shown in figure:



(*a*) $11\frac{2R}{3}$ (*b*) 14 R

(*c*) 21 R (*d*) $\frac{8}{3}$R

64. In the given circuit, the equivalent resistance between the terminal *A* and *B* is _____ Ω.



65. If you are provided a set of resistances 2Ω, 4Ω, 6Ω and 8Ω. Connect these resistances so as to obtain an equivalent 46 resistance of $\frac{46}{3}\Omega$

(*a*) 4 Ω and 6 Ω are in parallel with 2 Ω and 8 Ω in series.

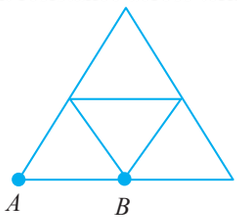
(*b*) 6 Ω and 8 Ω are in parallel with 2 Ω and 4 Ω in series.

(*c*) 2 Ω and 4 Ω are in parallel with 6 Ω and 8 Ω in series.

(*d*) 2 Ω and 6 Ω are in parallel with 4 Ω and 8 Ω in series.

66. A square shaped wire with resistance of each side 3 Ω is bent to form a complete circle. The resistance between two diametrically opposite points of the circle in unit of Ω will be

67. In the given diagram, resistance between any two junctions is R. Equivalent resistance across terminals A and B is



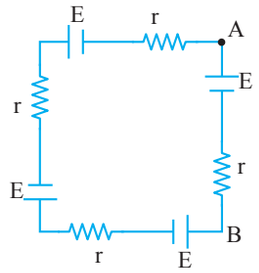
(*a*) $\frac{11R}{7}$ (*b*) $\frac{18R}{7}$ (*c*) $\frac{7R}{7}$ (*d*) $\frac{11R}{18}$

Kirchhoff Law and Circuit Analysis

68. **Assertion (A):** If potential difference between two points is zero, and the resistance between those points is zero, current may flow between the points.

Reason (R): Kirchhoff's first law is based on conservation of charge.

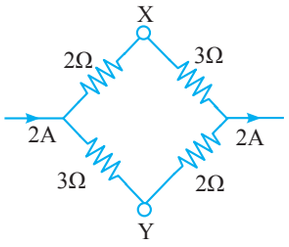
69. Potential difference across AB in the network shown in



(*a*) Zero (*b*) E

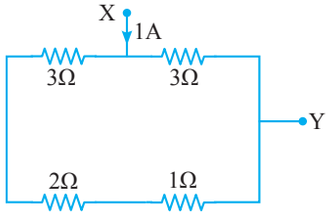
(*c*) $E-\frac{Ir}{2}$ (*d*) E – 2 *Ir*

70. Find $V_x - V_y = ?$



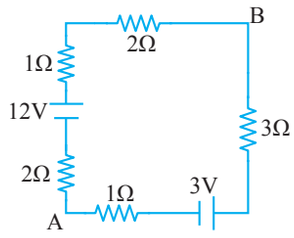
(*a*) 1 (*b*) –1 (*c*) 2 (*d*) –2

71. Find $V_x - V_y = ?$



(*a*) 2 (*b*) 3 (*c*) 6 (*d*) 9

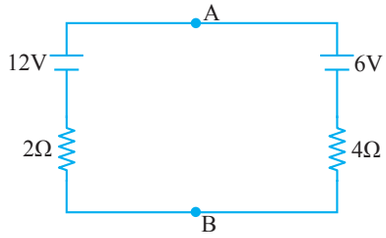
72. Potential difference $V_B - V_A$ in the network shown in



(*a*) 7 V (*b*) 6 V

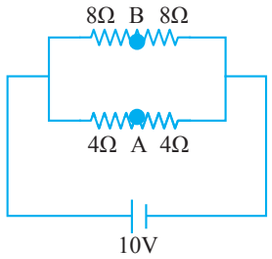
(*c*) 5 V (*d*) 8 V

73. Potential difference across AB *i.e.*, $V_A - V_B$ is



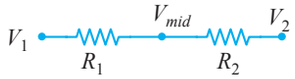
(*a*) 10 V (*b*) 8 V (*c*) 6 V (*d*) Zero

74. The potential difference between points A and B is

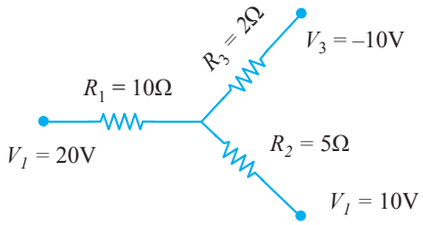


(*a*) $\frac{20}{7}$ V (*b*) $\frac{40}{7}$ V (*c*) $\frac{10}{7}V$ (*d*) Zero

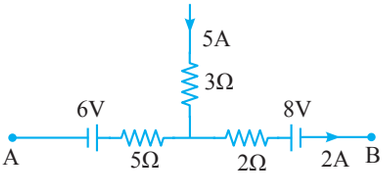
75. Find $V_{mid} = ??$.



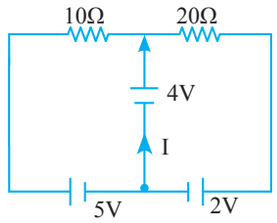
76. Find potential at junction.



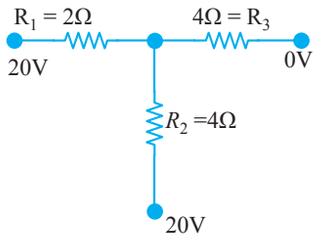
77. Find $V_B - V_A$ given circuit.



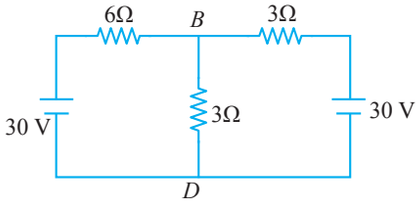
78. Current in 10Ω & 20Ω



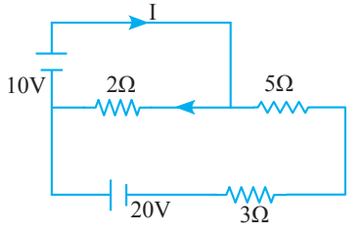
79. Find current in 2 Ω?



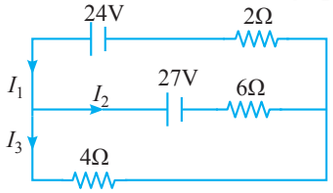
80. Find current in BD Branch.



81. Find current I



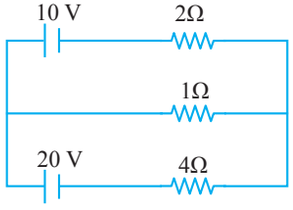
82. In given circuit, the value of currents I_1, I_2 and I_3 are



(*a*) $3A, \frac{-3}{2}A, \frac{9}{2}A$ (*b*) $\frac{9}{2}A, 3A, \frac{-3}{2}A$

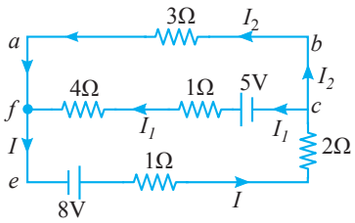
(*c*) $5A, 4A, -3A$ (*d*) $7A, \frac{5}{4}A, \frac{9}{2}A$

83. The value of current through 2 Ω resistor is

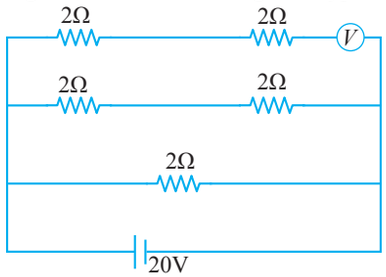


(*a*) 1.0 A (*b*) 1.5 A (*c*) 5.0 A (*d*) 2.1 A

84. In the network in given figure find I_1 , I_2 and I.



85. The reading of ideal voltmeter in given hypothetical circuit is

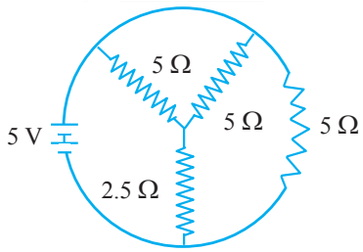


(*a*) 5 V (*b*) 20 V (*c*) 10 V (*d*) 16 V

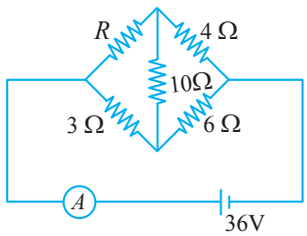
86. In a building there are 15 bulbs of 45 W, 15 bulbs of 100 W, 15 small fans of 10 W and 2 heaters of kW. The voltage of electric main is 220 V. The minimum fuse capacity (rated value) of the building will be

(*a*) 10 A (*c*) 25 A (*b*) 20 A (*d*) 15 A

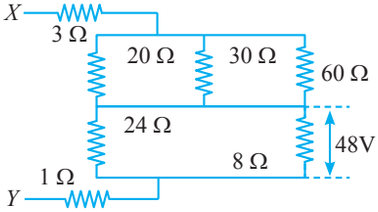
87. The total current supplied to the circuit as shown in figure by the 5 V battery is_____*A*.



88. Current measured by the ammeter (*A*) in the reported circuit when no current flows through 10 resistance, will be _____A.

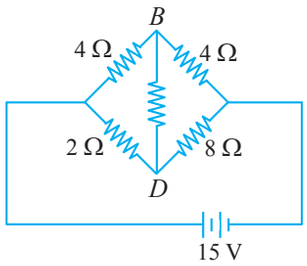


89. The potential difference across 8 ohm resistance is 48 volt, as shown in the figure. The value of potential difference across X and Y points will be



(*a*) 160 V (*b*) 128 V (*c*) 80 V (*d*) 62 V

90. In the circuit shown, if a conducting wire is connected between points B and D, then current in this wire will



(*a*) Flow from B to D
(*b*) Flow from D to B
(*c*) Be zero
(*d*) Data is insufficient to answer

Charging & Discharging of a Cell

91. A student measures the terminal potential differences (V) of cell (of emf and internal resistance r) as a function of the current (I) flowing through it. The slope, and intercept,of the graph between V and I, will respectively be equal to.

(*a*) $-r$ and ε (*b*) r and $-\varepsilon$
(*c*) $-\varepsilon$ and r (*d*) ε and $-r$

92. According to this diagram, the potential differences across the terminals is (internal resistance of cell = r)



(*a*) $V = E - Ir$ (*b*) V = E + *i r*
(*c*) V = E (*d*) Zero

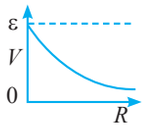
93. The internal resistance of a 2.1 V cell which gives a current of 0.2 A through a resistance of 10Ω is

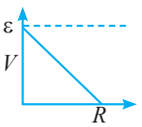
(*a*) 0.8 Ω (*b*) 1.0 Ω (*c*) 0.2 Ω (*d*) 0.5 Ω

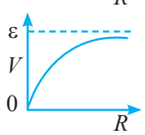
94. A current of 2 A flows through a 2Ω resistor when connected across a battery. The same battery supplies a current of 0.5 A when connected across a 9 Ω resistor. The internal resistance of the battery is.

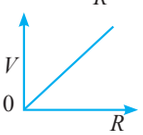
(*a*) 0.5 Ω (*b*) 1/3 Ω (*c*) 1/4 Ω (*d*) 1 Ω

95. A cell having an emf ε and internal resistance r is connected across a variable external resistance R. As the resistance R is increased, the plot of potential difference V across R is given by

(*a*) 

(*b*) 

(*c*) 

(*d*) 

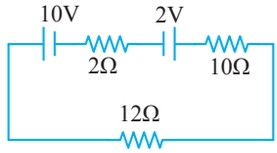
96. A car battery of emf 12V and internal resistance 5×10^{-2} Ω, receives a current of 60amp from external source, then terminal potential difference of battery is

(*a*) 12 V (*b*) 9 V (*c*) 15 V (*d*) 20 V

97. The internal resistance of a cell of e.m.f. 2V is 0.1Ω. It is connected to a resistance of 3.9 Ω. The voltage across the cell will be

(*a*) 1.95 V (*b*) 1.9 V
(*c*) 0.5 V (*d*) 2 V

98. Terminal potential across 10 V battery is x and across 2 volt battery is y. If $x + y = \frac{11A}{3}$, find the value of A.



99. Two cells, having the same e.m.f. Are connected in series through an external resistance R. Cells have internal resistances r_1 and r_2 respectively. When the circuit is closed, the potential difference across the first cell is zero. The value of R is

(*a*) $r_1 + r_2$ (*b*) $r_1 - r_2$
(*c*) $\frac{r_1 + r_2}{2}$ (*d*) $\frac{r_1 - r_2}{2}$

100. When a current of 2 A flows in a battery from negative to positive terminal, the potential difference across it is 12 V. If a current of 3 A flows in the opposite direction potential difference across the terminals of the battery is 15 V, the emf of the battery is

(*a*) 12.6 V (*b*) 13.2 V (*c*) 13.5 V (*d*) 14.0V

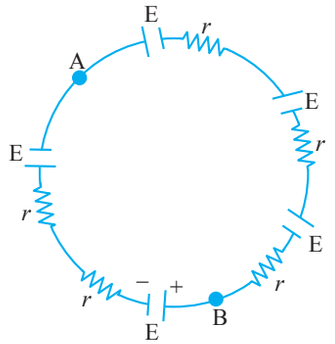
Series and Parallel Combination of Cell

101. If out of n identical cells m are wrongly connected then find equivalent emf & equivalent internal resistance?

102. Five cells each of e.m.f. E and internal resistance r are connected in series. If due to over sight, one cell is connected wrongly, then the equivalent e.m.f. of the combination is

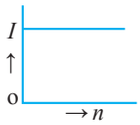
(*a*) 5 E (*b*) 2 E (*c*) 3 E (*d*) 4 E

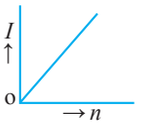
103. Find potential different between A & B

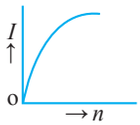


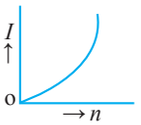
104. A battery consists of a variable number n of identical cells (having internal resistance r each) which are connected in

series. The terminals of the battery are short-circuited and the current I is measured. Which of the graphs shows the correct relationship between I and n?

(a) 

(b) 

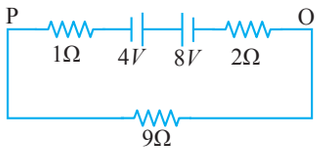
(c) 

(d) 

105. Thousand cells of same emf E and same internal resistance r are connected in series in same order without as external resistance. The potential drop across 399 cells is found to be

(a) Zero (b) 399 E (c) 601 E (d) 1000 E

106. Two batteries of emf 4 V and 8 V having the internal resistance of 1 Ω and 2 Ω respectively are connected in circuit with a resistance of 9 Ω as shown in figure. The current and potential difference between the points P and Q are



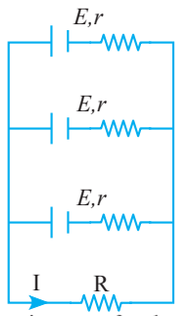
(a) $\frac{1}{2}$ A and 12 V (b) $\frac{1}{9}$ A and 9 V

(c) $\frac{1}{6}$ A and 4 V (d) $\frac{1}{3}$ A and 3 V

107. Two batteries of different e.m.f.'s and internal resistance connected in series with each other and with an external load resistor. The current is 3.0 A. When the polarity of one battery is reversed, the current becomes 1.0. A. The ratio of the e.m.f's of the two batteries is

(a) 2.5 (b) 2 (c) 1.5 (d) 1

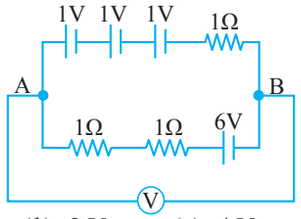
108. n-Identical battery connected in parallel with external resistance (R)then current in extenal resistance.



109. A set of 'n' equal resistors, of value 'R' each, are connected in series to a battery of emf 'E' and internal resistance 'R'. The current drawn is I. Now, the 'n' resistors are connected in parallel to the same battery. Then the current drawn from battery becomes 10 I. The value of 'n' is

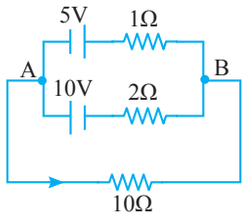
(a) 10 (b) 11 (c) 9 (d) 20

110. Reading of an ideal voltmeter in the circuit below is



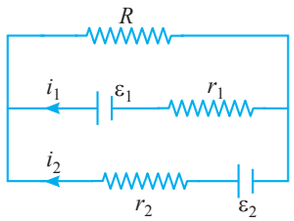
(a) zero (b) 2 V (c) 4 V (d) 6 V

111. Current through 10 Ω resistor shown in figure is



(a) zero (b) 1 A (c) 1.5 A (d) 2 A

112. See the electrical circuit shown in this figure. Which of the following equations is a correct equation for it?



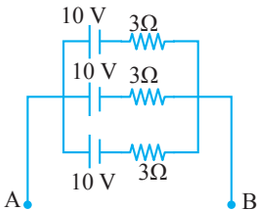
(a) $\varepsilon_2 - i_2 r_2 - \varepsilon_1 - i_1 r_1 = 0$

(b) $-\varepsilon_2 - (i_1 + i_2) R + i_2 r_2 = 0$

(c) $\varepsilon_1 - (i_1 + i_2) R + i_1 r_1 = 0$

(d) $\varepsilon_1 - (i_1 + i_2) R - i_1 r_1 = 0$

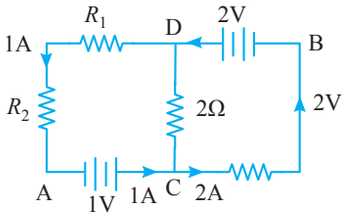
113. Three identical cells are connected in parallel across AB. Net emf across AB is.



(a) 10 V (b) 30 V

(c) 15 V (d) 12 V

114. In the circuit shown in the figure, if the potential at point A is taken to be zero, the potential at point B is



(a) +1 V (b) –1 V

(c) +2 V (d) –2 V

115. The combination of two identical cells, whether connected in series or parallel combination provides same current through external resistor of 2Ω resistance. Value of internal resistance of each cell is :

(*a*) 2 Ω (*b*) 4 Ω (*c*) 6 Ω (*d*) 8 Ω

116. *n* identical cell [ε, *r*] connected in series and this arrangement is in *m*-times parallel. Current I through load resistance *R* will be maximum when

(*a*) $R = \frac{nr}{m}$ (*b*) $R = \frac{m}{nr}$

(*c*) $R = r$ (*d*) $R = nr$

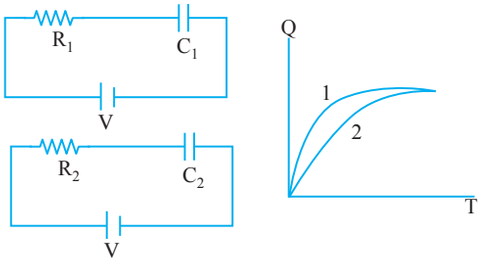
117. There are a large number of cells available, each marked (6 V, 0.5 Ω) to be used to supply current to a device of resistance 0.75 Ω, requiring 24 A current. How should the cells be arranged, so that power is transmitted to the load using minimum number of cells?

(*a*) Six rows, each containing four cells
(*b*) four rows, each containing six cells
(*c*) Four rows, each containing four cells
(*d*) Six rows, each containing six cells

Charging of Capacitor

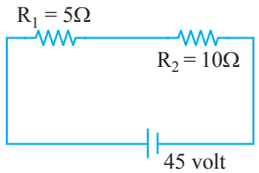
118. Find time when charge on capacitor is half of the maximum charge.

119. Compare C_1, C_2, and R_1, R_2 from charge – time graph for both the circuit

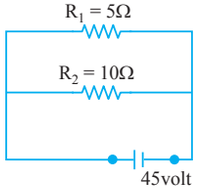


Power Loss

120. Find ratio of Power loss in 5 Ω and 10 Ω and also find total power loss in circuit.



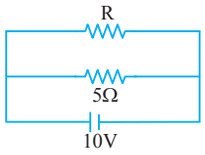
121. Find ratio of power loss in 5 Ω and 10 Ω and also find total power loss in circuit?



122. Kilowatthour is the unit of

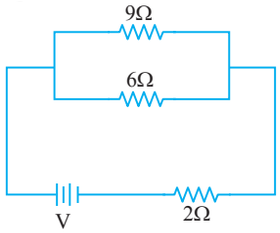
(*a*) Energy (*b*) Power (*c*) Charge (*d*) Time

123. The power dissipated in the circuit shown in the figure is 30 watts. The value of R is



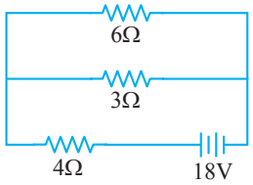
(*a*) 20 Ω (*b*) 15 Ω (*c*) 10 Ω (*d*) 30 Ω

124. If power dissipated in the 9 Ω resistor in the circuit shown is 36 watt, the potential difference across the 2 Ω resistor is



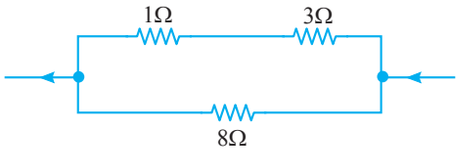
(*a*) 4 volt (*b*) 8 volt (*c*) 10 volt (*d*) 2 volt

125. The total power dissipated in watt in the circuit shown here is



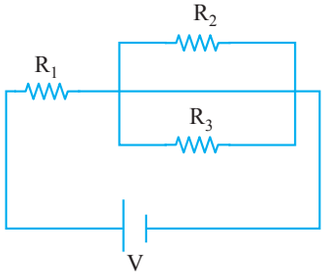
(*a*) 40 (*b*) 54 (*c*) 4 (*d*) 16

126. Power dissipated across the 8 Ω resistor in the circuit shown here is 2 watt. The power dissipated in watt units across the 3 Ω resistor is



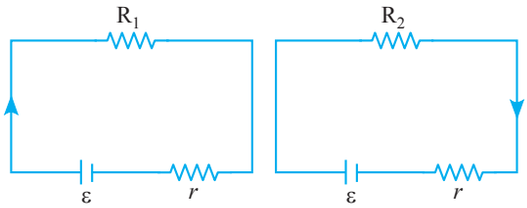
(*a*) 3.0 (*b*) 2.0 (*c*) 1.0 (*d*) 0.5

127. Three identical resistors $R_1 = R_2 = R_3$ are connected as shown to a battery of constant e.m.f. The power dissipated is



(*a*) The least in R_1
(*b*) Greatest in R_1
(*c*) In the ratio 1 : 2 in resistance R_1 and R_2 respectively
(*d*) The same in R_1 and in the parallel combination of R_2 and R_3

128. Two unequal resistances R_1 and R_2 are connected across two identical batteries of emf ε and internal resistance r. Can the thermal energy developed in R_1 and R_2 be equal. If yes, find the condition.



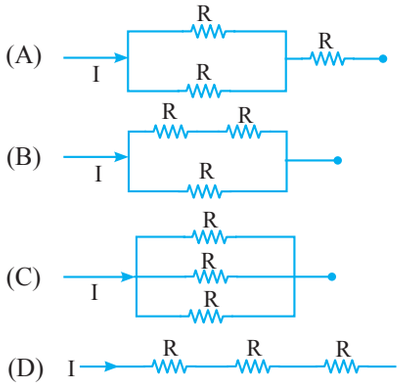
129. Three equal resistors connected in series across a source of EMF together dissipate 10 W of power. What would be the power dissipate, if the same resistors are connected in parallel across the same source of EMF?

(*a*) 25 W (*b*) 80 W (*c*) 45 W (*d*) 90 W

130. Four equal resistance dissipated 5 W of power together when connected in series to a battery of negligible internal resistance. The total power dissipated in these resistance when connected in parallel across the same battery would be

(*a*) 125 W (*b*) 80 W (*c*) 20 W (*d*) 5 W

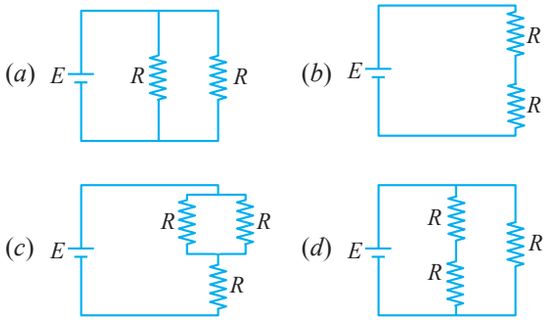
131. Different combination of 3 resistors of equal resistance R are shown in the figures. The increasing order for power dissipation is :



(*a*) $P_B < P_C < P_D < P_A$ (*b*) $P_C < P_B < P_A < P_D$

(*c*) $P_C < P_D < P_A < P_B$ (*d*) $P_A < P_B < P_C < P_D$

132. Consider the four circuits, shown in the figure. In which circuit the power dissipated is the greatest? (Neglect the internal resistance of the power supply)



133. Consider two identical cells each Of emf E and internal resistance r connected to a load resistance R.

	Column-I		Column-II
A.	Maximum power transfer to load if cells are connected in series	(i)	$\frac{E^2}{4r}$
B.	Maximum power transfer to Load if cells are connected in parallel	(ii)	$\frac{E^2}{2r}$
C.	For series combination of cells	(iii)	$E_{eq} = E, r_q = \frac{r}{2}$
D.	For parallel connected of cells	(iv)	$E_{eq} = 2E, r_{eq} = 2r$

(*a*) A-(ii), B-(i), C-(iii), D-(iv)

(*b*) A-(i), B-(iii), C-(ii), D-(iv)

(*c*) A-(ii), B-(ii), C-(iv), D-(iii)

(*d*) A-(i), B-(iv), C-(ii), D-(iii)

Joule's Law of Heating

134. Current 2 A is flowing through a conductor of resistance 4Ω. Find the electrical energy consumed in 10 s.

135. The charge flowing through a resistance R varies with time t as $Q = at - bt^2$, where a and b are positive constants. The total heat produced in R is

(*a*) $\frac{a^3R}{b}$ (*b*) $\frac{a^3R}{6b}$ (*c*) $\frac{a^3R}{3b}$ (*d*) $\frac{a^3R}{2b}$

136. An electric kettle takes 4 A current at 220 V. How much time will it take to boil 1 kg of water, from temperature 20°*C*? (The temperature of boiling water is 100°*C*.)

(*a*) 4.2 min (*b*) 6.3 min

(*c*) 8.4 min (*d*) 12.6 min

137. To drive a device of power P, the power wasted in the connecting wires is inversely proportional to

Bulb

138. Two bulbs are marked (*a*) 100 W, 220 V (*b*) 40 W, 220 V. Which has higher resistance ?

139. An electric bulb marked 40 W and 200 V , is used in a circuit of supply voltage 100 V. Its power consumed

(*a*) 10 W (*b*) 20 W

(*c*) 40 W (*d*) 100 W

140. If voltage across a bulb rated 220 V – 100 W drops by 2.5 % of its rated value, the percentage of the rated value by which the power would decrease is

(*a*) 5% (*b*) 10% (*c*) 20% (*d*) 2.5%

141. A current through a lamp marked 50 W, 250 V is

(*a*) 0.1 A (*b*) 0.2 A (*c*) 0.4 A (*d*) 0.3 A

142. An electric bulb of 500 watt at 100 volt is used in circuit having a 200 V supply. Calculate resistance R to be connected in series with bulb so that power delivered by bulb is 500 W .

(*a*) 20 Ω (*b*) 5 Ω (*c*) 30 Ω (*d*) 10 Ω

143. **Assertion (A):** When current through a bulb is increased by 2%, power increases by 4%.

Reason (R): Current passing through the bulb is inversely proportional to its resistance.

Series & Parallel Combination of Bulbs

144. Three bulbs of 40 W, 60 W and 100 W are arranged in series with 220 V. Which bulb has minimum resistance?

(*a*) 100 W (*b*) 40 W

(*c*) 60 W (*d*) Equal in all bulbs

145. Rated individual power of two elements are 100 W and 150 W. Effective power of their series combination will be

(*a*) 120 W (*b*) 75 W (*c*) 50 W (*d*) 60 W

146. Two electric bulbs whose resistances are in the ratio 1 : 2 are connected in series to a constant voltage source. The power dissipated in them are in the ratio

(*a*) 1 : 2 (*b*) 1 : 1 (*c*) 1 : 4 (*d*) 2 : 1

147. When three identical bulbs of 60 watt, 200 volt rating are connected in series to a 200 volt supply, the power drawn by them will be

(*a*) 1.60 watt (*b*) 180 watt

(*c*) 10 watt (*d*) 20 watt

148. Two electric bulbs A and B rated 200 V ~ 100 W and 200 V ~ 60 W are connected in series to a 200 V line. Then the potential drop across

(*a*) Each bulb is 200 V

(*b*) 100 W bulb is greater than that across 60 W bulb

(*c*) 100 W bulb is smaller than that across 60 W bulb

(*d*) Each bulb is 100 V

149. Two (220 volt, 100 watt bulbs) are connected first in series and then in parallel. Each time the combination is connected to a 220 volt a.c. supply line. The power drawn by the combination in each case respectively will be

(*a*) 50 watt, 100 watt (*b*) 100 watt, 50 watt

(*c*) 200 watt, 150 watt (*d*) 50 watt, 200 watt

150. Of the two bulbs in a house hold circuit one glows brighter than the other. Which of the two bulbs has a large resistance?

(*a*) The bright bulb

(*b*) The dim bulb

(*c*) Both have the same resistance

(*d*) The brightness does not depend upon the resistance

151. If two bulbs of power 60 W and 100 W respectively each rated 110 V are connected in series with the supply of 220 V, then

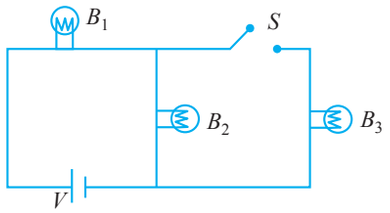
(*a*) 60 W bulb will fuse

(*b*) 100 W bulb will fuse

(*c*) Both bulbs will fuse

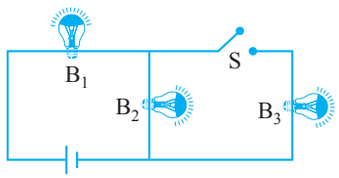
(*d*) Bulbs will not fuse

152. Three identical bulbs B_1, B_2 and B_3 are connected to the mains as shown in figure. If B_3 is disconnected from the circuit by opening switch *S*, then incandescence of bulb B_1 will



(*a*) increase (*b*) decrease (*c*) constant (*d*) can't say

153. Three identical bulbs B_1, B_2, B_3 are connected to mains as shown in figure. If B_3 is disconnected from circuit by opening switch *S*, then incandescence of bulb B_2 will



(*a*) Increase (*b*) Decrease

(*c*) Become zero (*d*) No change

154. How many 60 W bulbs may be safely run on 220 V using a 5 A fuse?

(*a*) 18 (*b*) 16 (*c*) 14 (*d*) 12

Series & Parallel Combination of Heating Coils

155. A heater coil connected across a given potential difference has power P. Now, the coil is cut into two equal halves and joined in parallel. Across the same potential difference, this combination has power

(*a*) P (*b*) 4P (*c*) P/4 (*d*) 2P

156. Two identical heater filaments are connected first in parallel and then in series. At same applied voltage, ratio of heat produced in same time for parallel to series will be:

(*a*) 4 : 1 (*b*) 1 : 4 (*c*) 2 : 1 (*d*) 1 : 2

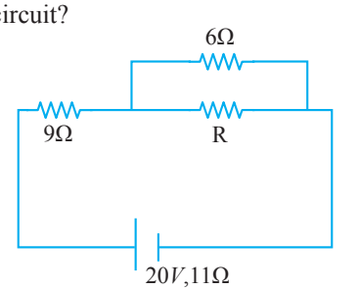
157. A heater boils certain amount of water in 15 minute. Another heater boils same amount of water in 10 minute. Time taken to boil same amount of water when both are used in parallel is

(*a*) 25 minute (*b*) 6 minute

(*c*) 12 minute (*d*) 12.5 minute

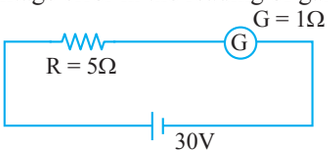
Maximum Power Theorem

158. Find R so that battery will supply maximum power to external circuit?

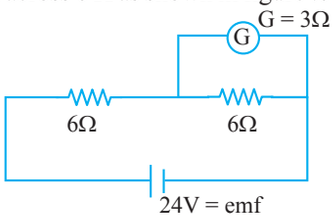


Galvanometer

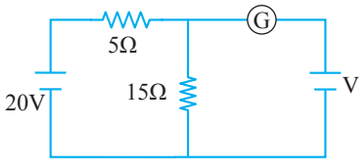
159. Find percentage error in the reading of galvanometer.



160. Find percentage error in the reading of galvanometer connected across 6 Ω as shown in figure to measure voltage.



161. If current in galvanometer is Zero; then find Value of V:



Ammeter and Voltmeter

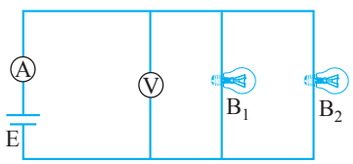
162. Two identical bulbs are connected in parallel across an ideal source of emf E. The ammeter A and voltmeter V are ideal. If bulb B_2 gets fused, then

(*a*) Reading of A will increase but that of V will remain same

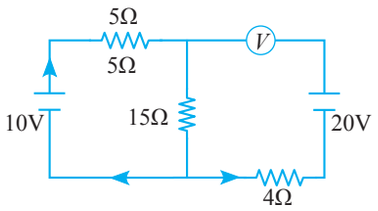
(*b*) Reading of A will decrease but that of V will increase

(*c*) Reading of A will decrease but that of V will remain same

(*d*) Reading of A will increase and reading of V will also increase



163. Find reading of voltmeter



164. **Assertion (A):** For measuring the potential difference across a resistance of 600 Ω. the voltmeter with resistance 1000 Ω will be preferred over voltmeter with resistance 4000 Ω

Reason (R): Voltmeter with higher resistance will draw smaller current than voltmeter with lower resistance.

(*a*) Assertion (A) is false but Reason (R) is trug

(*b*) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(*c*) Assertion (A) is true but Reason (R) is false

(*d*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

165. A circuit to verify Ohm's law uses ammeter and voltmeter in series or parallel connected correctly to the resistor. In the circuit:

(*a*) ammeter is always used in parallel and voltmeter is series.

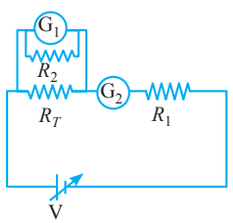
(*b*) both ammeter and voltmeter must be connected in parallel.

(*c*) ammeter is always connected in series and voltmeter in parallel.

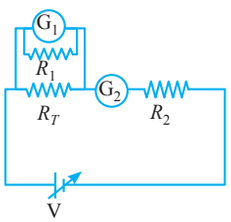
(*d*) both, ammeter and voltmeter must be connected in series.

166. A student is provided with a variable voltage source V, a test resistor R_r = 10Ω, two identical galvanometers G_1 and G_2 and two additional resistors, R_1 = 10MΩ and R_2 = 0.001Ω . For conducting an experiment to verify ohms law, the most suitable circuit is:

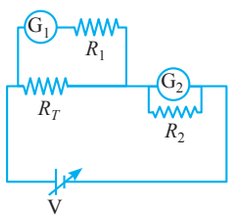
(*a*)



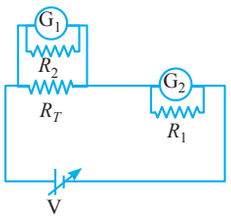
(*b*)



(*c*)



(*d*)



Galvanometer into Ammeter

167. To convert a galvanometer into a ammeter, one needs to connect a

(*a*) low resistance in parallel(*b*) high resistance in parallel

(*c*) low resistance in series (*d*) high resistance in series.

168. A galvanometer having a coil of resistance 60 Ω shows full scale deflection when a current of 1.0 amp passes through it. It can be converted into an ammeter to read currents upto 5.0 amp by

(*a*) putting in series a resistance of 15 Ω

(*b*) putting in series a resistance of 240 Ω

(*c*) putting in parallel a resistance of 15 Ω

(*d*) putting in parallel a resistance of 240 Ω

169. The resistance of an ammeter is 13 Ω and its scale is graduated for a current upto 100 amps. After an additional shunt has been connected to this ammeter it becomes possible to measure currents unto 750 amperes by this meter. The value of shunt resistance is

(*a*) 2 Ω (*b*) 0.2 Ω (*c*) 0.02 Ω (*d*) 20 Ω

170. Ammeter of resistance 100 Ω is used to measure 10 A. Find resistance of shunt if we want to measure 50 A .

171. A galvanometer having a resistance of 9 ohm is shunted by a wire of resistance 2 ohm . If the total current is 1 amp , the part of it passing through the shunt will be

(*a*) 0.2 amp (*b*) 0.8 amp (*c*) 0.25 amp (*d*) 0.5 amp

172. In an ammeter 0.2 % of main current passes through the galvanometer. If resistance of galvanometer is G, the resistance of ammeter will be

(*a*) $\frac{1}{499}$ G (*b*) $\frac{499}{500}$ G (*c*) $\frac{1}{500}$ G (*d*) $\frac{500}{499}$ G

173. A millivoltmeter of 25 millivolt range is to be converted into an ammeter of 25 ampere range. The value (in ohm) of necessary shunt will be

(*a*) 1 (*b*) 0.05

(*c*) 0.001 (*d*) 0.01

Galvanometer into voltmeter

174. The convert a galvanometer into a voltmeter one should connect a

(*a*) high resistance in series with galvanometer

(*b*) low resistance in series with galvanometer

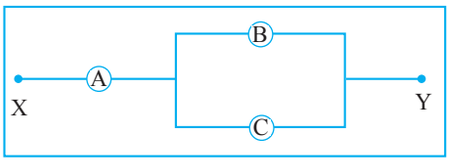
(*c*) high resistance in parallel with galvanometer

(*d*) high resistance in parallel with galvanometer

175. A, B and C are voltmeters of resistance R, 1.5 R and 3 R respectively as shown in the figure. When some potential difference is applied between X and Y, the voltmeter readings are V_A, V_B and V_C respectively, then

(*a*) $V_A \neq V_B \neq V_C$ (*b*) $V_A = V_B = V_C$

(*c*) $V_A \neq V_B = V_C$ (*d*) $V_A = V_B > V_C$



176. A galvanometer of resistance, G is shunted by a resistance S.ohm. To keep the main current in the circuit unchanged, the resistance to be put in series with the galvanometer is

(*a*) $\frac{G^2}{(S+G)}$ (*b*) $\frac{G}{(S+G)}$

(*c*) $\frac{G^2}{(S+2G)}$ (*d*) $\frac{SG}{(S+G)}$

177. A galvanometer has a coil of resistance 100 Ω and gives a full scale deflection for 30 mA current. (if it is to work as a voltmeter of 30 V range, the resistance required to be added will be

(*a*) 1000 Ω (*b*) 900 Ω

(*c*) 1800 Ω (*d*) 500 Ω

178. A galvanometer of resistance 50 Ω is connected to a battery of 3 V along with a resistance of 2950 Ω in series. A full scale deflection of 30 divisions is obtained in the galvanometer, In order to reduce, this deflection to 20 divisions, the resistance in series should be

(*a*) 4450 Ω (*b*) 5050 Ω

(*c*) 5550 Ω (*d*) 6050 Ω

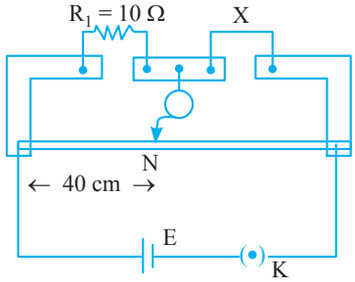
179. A galvanometer of 50 ohm resistance has 25 divisions. A current of 4×10^{-4} ampere gives a deflection of one division. To convert this galvanometer into a voltmeter having a range of 25 volts, it should be connected with a resistance of

(*a*) 2500 Ω as a shunt (*b*) 2450 Ω as a shunt

(*c*) 2550 Ω in series (*d*) 2450 Ω in series

Meter Bridge

180. In the meter bridge experiment, the null point is obtained at N . The value of unknown resistance X will



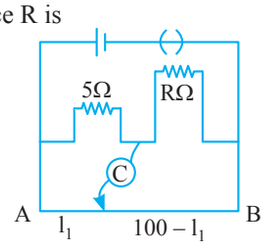
(*a*) 60 Ω (*b*) 40 Ω (*c*) 6 Ω (*d*) 15 Ω

181. A resistance wire connected in the left gap of a meter bridge balances a 10 Ω resistance in the right gap at a point which divides the bridge wire in the ratio 3 : 2. If the length of the resistance wire is 1.5 m , then the length of 1 Ω of the resistance wire is

(*a*) 1.0×10^{-2} m (*b*) 1.0×10^{-1}m

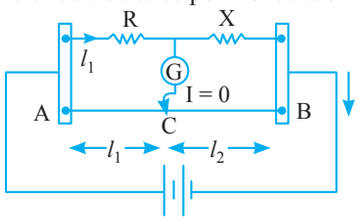
(*c*) 1.5×10^{-1} m (*d*) 1.5×10^{-2} m

182. The resistances in the two arms of the meter bridge are 5 Ω and R Ω respectively. When the resistance R is shunted with an equal resistance, the new balance point is at 1.6 l_1. The resistance R is



(*a*) 10 Ω (*b*) 15 Ω (*c*) 20 Ω (*d*) 25 Ω

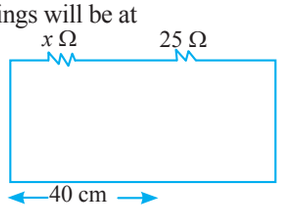
183. In the meter bridge shown, the resistance X has a negative temperature coefficient of resistance. Neglecting the variation in other resistor, when current is passed for some time, in the circuit balance point should shift towards



(*a*) A (*b*) B

(*c*) First A then B (*d*) It will remain at C

184. The resistance per centimeter of meter bridge wire is *r* with *x* Ω resistor in left gap. Balancing length from left end is a 40 cm with 25 Ω resistance in right gap. Now wire is replaced by another wire of 2*r* resistance per cm. New balancing length for same settings will be at



(*a*) 10 cm (*b*) 80 cm (*c*) 20 cm (*d*) 40 cm

Potentiometer Wire

185. Two cells of emf E_1 and E_2 are to be compared in a potentiometer ($E_1 > E_2$). When the cells are used in series correctly, the balancing length obtained is 400 cm , When they are used in series but E_2 is connected with reverse polarities, the balancing length obtained is 200 cm . Ratio of emf of cells is

(*a*) 3 : 2 (*b*) 3 : 1 (*c*) 4 : 1 (*d*) 4 : 3

186. Two cells of e.m.f. E_1 and E_2 are joined in series and the balancing length of the potentiometer wire is 625 cm . If the terminals of E_1 are reversed, the balancing length obtained is 125 cm . Given $E_2 > E_1$ the ratio $E_1 : E_2$ will be

(*a*) 2 : 3 (*b*) 5 : 1 (*c*) 3 : 2 (*d*) 1 : 5

187. A potentiometer wire is 100 cm long and a constant potential difference is maintained across it Two cells are connected in series first to support one another and then in opposite direction. The balance points are obtained at 50 cm and 10 cm from the positive end of the wire in the two cases. The ratio of emf's is

(*a*) 3 : 2 (*b*) 5 : 1 (*c*) 5 : 4 (*d*) 3 : 4

188. A 10 m long potentiometer wire is connected to a battery having a steady voltage. A Leclanche cell is balanced at 4 m length of the wire. If the length is kept the same, but its cross-section is doubled, the null point will be obtained at

(*a*) 8 m (*b*) 4 m

(*c*) 2 m (*d*) None of these

189. A potentiometer wire of length *L* and a resistance *r* are connected in series with a battery of e.m.f. E_0 and a resistance r_1. An unknown e.m.f. *E* is balanced at a length *l* of the potentiometer wire. The e.m.f. *E* will be given by

(*a*) $\frac{LE_0 r}{(r+r_1)l}$ (*b*) $\frac{LE_0 r}{lr_1}$ (*c*) $\frac{E_0 r}{(r+r_1)}\frac{l}{L}$ (*d*) $\frac{E_0}{L}$

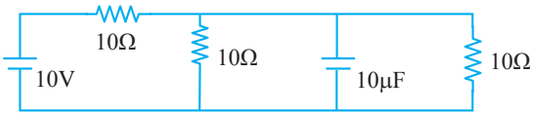
190. A potentiometer circuit has been set up for finding the internal resistance of given cell. The main battery, used across the potentiometer wire, has an emf of 2.0 V and a negligible internal resistance. The potentiometer wire itself is 4m long. When the resistance, *R*, connected across the given cell, has values of (*i*) Infinity, (*ii*) 9.5 Ω, the 'balancing lengths', on the potentiometer wire are found to be 3 m and 2.85 m, respectively. The value of internal resistance of the cell is

(*a*) 0.25 Ω (*b*) 0.95 Ω (*c*) 0.5 Ω (*d*) 0.75 Ω

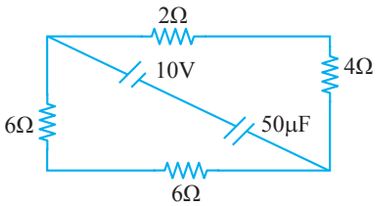
DC - RC Circuit

191. (*i*) Find *I* at $t = 0$ when Key is closed

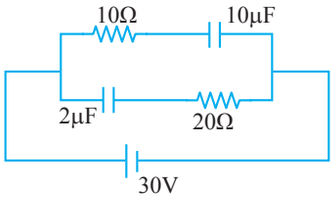
(*ii*) Find *I* at $t = \infty$ from battery (steady State)



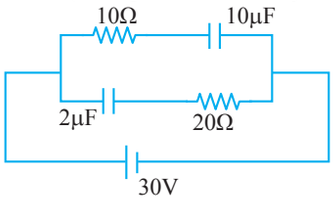
192. Find current – at $t = 0$ and at $t = \infty$.



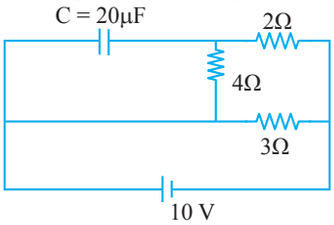
193. Find current from Battery at $t = 0$, and at steady state.



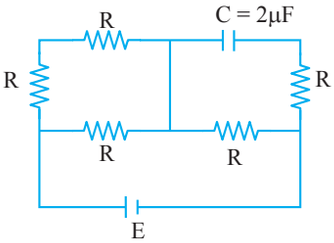
194. Find charge on 2 μF capacitor of steady state



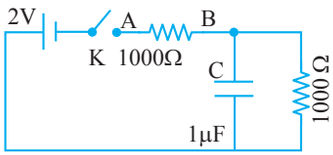
195. Find charge on capacitor at steady state.?



196. Find charge on capacitor at steady state



197. When the key K is pressed at time $t = 0$, then which of the following statement about the current I in the resistance AB of resistance 1000 Ω of the given circuit is true?



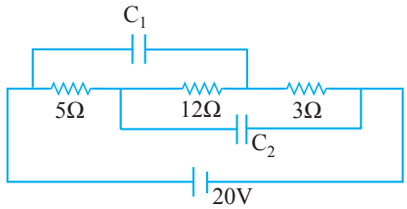
(*a*) I oscillates between 1 mA and 2 mA

(*b*) At $t = 0$, $I = 2$ mA and with time it goes to 1 mA

(*c*) $I = 1$ mA at all t

(*d*) $I = 2$ mA at all t

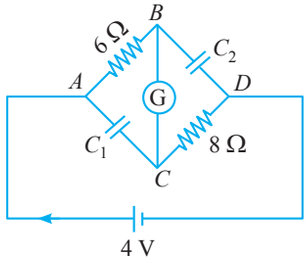
198. At steady state find Potential difference across C_1 & C_2



(*a*) 17 V, 3 V (*b*) 15 V, 3V

(*c*) 17 V, 15 V (*d*) 5V, 3V

199. In this figure the resistance of the coil of galvanometer G is 2 Ω. The emf of the cell is 4 V. The ratio of potential difference across C_1 and C_2 is:

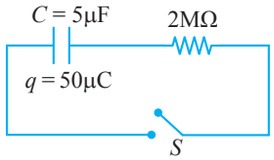


(*a*) 1 (*b*) $\frac{4}{5}$

(*c*) $\frac{3}{4}$ (*d*) $\frac{5}{4}$

Discharging of Capacitor

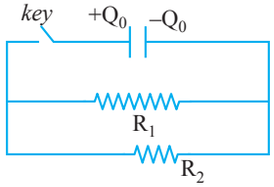
200. The following circuit consists of a 5 μF capacitor, having charge 50 μ C as shown. The switch is closed at $t = 0$. The value of current in 2 MΩ resistor at $t = 0$ is



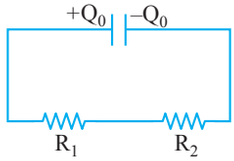
(*a*) 1 μA (*b*) 2 μA

(*c*) 5 μA (*d*) 5 A

201. Find Ratio of heat loss in R_1 and R_2. When capacitor is discharging through R_1 & R_2



202. Find ratio of heat loss across R_1 and R_2 when capacitor is discharging.



203. Find time when energy on capacitor is half of the maximum energy.

MR* CORNER

1. **Statement I:** A metallic resistor is connected across a battery. If the number of collisions of free electron with the lattice is somehow decreased in the resistor, the current will increase.

   Statement II: The product of resistivity and conductivity of a cylindrical conductor depends on the material of conductor.

2. **Statement I:** As the temperature of metallic resistor is increased the product of its resistivity & conductivity increases.

   Statement II: In an electric circuit containing a battery the charge (assumed positive) inside the battery always goes from negative terminal to the positive terminal.

3. **Statement I:** A resistor of resistance R is connected to an ideal battery. If the value of R is decreased. The power dissipated in resistor will increase.

   Statement II: Two resistors R and 2R are connected in series in an electric circuit. The Thermal energy developed in R and 2R are in Ratio 2 : 1

4. **Statement I:** A uniform wire of resistance 50Ω is cut longitudinally into 5 parts each having same area cross section but length being same as their parent wire. These parts are now connected in parallel. The equivalent resistance of combination will Remain 50Ω.

   Statement II: Kirchhoff's loop law follows from conservative nature of electric field.

5. **Statement I:** Two non-ideal batteries are connected in series. The equivalent emf is longer than either of the Two emfs.

   Statement II: Two non-ideal batteries are connected in parallel. The equivalent emf is smaller than either of the two emfs.

6. **Statement I:** Two non-ideal batteries are connected in series. The equivalent internal resistance is longer than either of the Two internal resistances.

   Statement II: Two non-ideal batteries are connected in parallel. The equivalent internal resistance is smaller than either of the two internal resistance.

7. **Statement I:** The net resistance of the ammeter should be small to ensure that it does not appreciably change the current to be measured.

   Statement II: The net resistance of voltmeter should be large to ensure that it does not get overheated.

8. **Statement I:** When no current is passed through a conductor, the free electrons do not move.

   Statement II: When no current is passed through a conductor, the average speed of free electron over a large period of time is zero.

9. **Statement I:** Number of free electrons and resistivity of resistor do not changes when resistor connected to a battery is heated due to the current.

   Statement II: As the Temperature of conductor increases its ratio of resistivity to conductivity increases.

10. **Statement I:** A current passes through a wire of non-uniform Cross-section. The charge crossing in a given interval of time. Remain independent of the cross-section.

    Statement II: A current passes through a wire of non-uniform cross-section. The current density remains independent of the cross-section.

Answer Key

1. (*a*) **2.** (*c*) **3.** (2A) **4.** (*c*) **5.** 6A, 4A **6.** (*d*) **7.** (3 : 1) **8.** (*a*) **9.** $\frac{25}{4}A$ **10.** $\frac{10}{8}A$

11. 2 times **12.** (*b*) **13.** (*a*) **14.** (*c*) **15.** (*d*) **16.** (*d*) **17.** (*a*) **18.** (*b*) **19.** (*a*) **20.** (*a*)

21. (*b*) **22.** (*a*) **23.** (No) **24.** $\mu_e >> \mu_p$ **25.** (4) **26.** (*b*) **27.** (*a*) **28.** (*c*) **29.** (*d*) **30.** (*b*)

31. (*c*) **32.** (*a*) **33.** (*b*) **34.** (*b*) **35.** (*c*) **36.** (*d*) **37.** (*d*) **38.** (*b*) **39.** (*c*) **40.** (*a*)

41. (*b*) **42.** (*a*) **43.** (*c*) **44.** (*d*) **45.** (*a*) **46.** $V_B = -11$ volt **47.** (*a*) **48.** (*c*)

49. 10V, 60V, 30V **50.** $I_{20\Omega} = 1A, I_{5\Omega} = 4A$ **51.** (*c*) **52.** (*c*) **53.** (*a*) **54.** $\frac{2E}{3}V$ **55.** (*c*) **56.** (*b*)

57. (*b*) **58.** (*b*) **59.** (*b*) **60.** 5A, 15V **61.** (*b*) **62.** (*a*) **63.** (*d*) **64.** 10 Ω **65.** (*c*) **66.** 3Ω

67. (*d*) **68.** (*b*) **69.** (*a*) **70.** (*a*) **71.** (*a*) **72.** (*a*) **73.** (*a*) **74.** (*d*) **75.** $\frac{\left(\frac{V_1}{R_1}+\frac{V_2}{R_2}\right)}{\left(\frac{1}{R_1}+\frac{1}{R_1}\right)}$

76. $\frac{-10}{8}V$ **77.** 9V **78.** $I_{10\Omega} = 0.1A, I_{20\Omega} = 0.1A$ **79.** $\frac{5}{2}A$ **80.** 6A **81.** $\frac{35}{4}A$ **82.** (*a*) **83.** (*d*)

84. $I_1 = \frac{18}{13}A, I_2 = \frac{25}{39}A, I_3 = \frac{79}{39}A$ **85.** (*b*) **86.** (*b*) **87.** (2) **88.** 10A **89.** (*a*) **90.** (*b*)

91. (*a*) **92.** (*b*) **93.** (*d*) **94.** (*b*) **95.** (*c*) **96.** (*c*) **97.** (*a*) **98.** (4) **99.** (*b*) **100.** (*b*)

101. (*nE* –2*mE*), *nr* **102.** (*c*) **103.** (0) **104.** (*a*) **105.** (*a*) **106.** (*d*) **107.** (*b*) **108.** $I = \frac{3E}{3R+r}$

109. (*a*) **110.** (*c*) **111.** (*a*) **112.** (*d*) **113.** (*a*) **114.** (*a*) **115.** (*a*) **116.** (*a*) **117.** (*b*)

118. $t = RC \ln 2 = \tau \ln 2$ **119.** $C_1 = C_2, R_1 < R_2$ **120.** 1 : 2, 135 W **121.** 2 : 1, 607.5 W **122.** (*a*) **123.** (*c*)

124. (*c*) **125.** (*b*) **126.** (*a*) **127.** (*b*) **128.** $\left(\text{Yes}, r = \sqrt{R_1 R_2}\right)$ **129.** (*d*) **130.** (*b*) **131.** (*b*) **132.** (*a*)

133. (*c*) **134.** 160 J **135.** (*b*) **136.** (*b*) **137.** Square of voltage across the device **138.** $R_{100W} < R_{40W}$ **139.** (*a*)

140. (*a*) **141.** (*b*) **142.** (*a*) **143.** (*b*) **144.** (*a*) **145.** (*d*) **146.** (*a*) **147.** (*d*) **148.** (*c*) **149.** (*d*)

150. (*b*) **151.** (*a*) **152.** (*b*) **153.** (*a*) **154.** (*a*) **155.** (*b*) **156.** (*a*) **157.** (*b*) **158.** (R = 3Ω)

159. (16.7%) **160.** 50% **161.** 15 volt **162.** (*c*) **163.** 12.5 volt **164.** (*a*) **165.** (*c*)

166. (*c*) **167.** (*a*) **168.** (*c*) **169.** (*a*) **170.** 25Ω **171.** (*b*) **172.** (*c*) **173.** (*c*) **174.** (*a*) **175.** (*b*)

176. (*a*) **177.** (*b*) **178.** (*a*) **179.** (*d*) **180.** (*d*) **181.** (*b*) **182.** (*b*) **183.** (*b*) **184.** (*d*) **185.** (*b*)

186. (*a*) **187.** (*a*) **188.** (*b*) **189.** (*c*) **190.** (*c*) **191.** $I_{(t=0)} = 1A, I_{(t=\infty)} = \frac{2}{3}A$

192. at $t = 0$, I = 2.5 A, at t = ∞, I = 0 **193.** at $t = 0$, I = 4.5 A, at t = ∞, I = 0 **194.** 60μC **195.** $\frac{400}{3}C$ **196.** $\frac{6E}{5}$ **197.** (*b*)

198. (*c*) **199.** (*b*) **200.** (*c*) **201.** $\frac{E_1}{E_2} = \frac{R_2}{R_1}$ **202.** $\frac{H_1}{H_2} = \frac{R_1}{R_2}$ **203.** $t = -RC \ln\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$

MR* CORNER

1. I. True, II. - False **2.** Both are False **3.** I. - True, II. - False **4.** Both are True.

5. I. - True, II. - False **6.** Both True **7.** I. - True, II. - False **8.** Both are False

9. I. - False, II. - True **10.** I. - True, II. - False